

# Probabilistic Machine Learning

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## Lecture 01: Introduction

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# Why Probability?

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**Guiding Question:** Two models both say "fraud". What would you need to know before acting on either — and why does a label not tell you?

# Machine Learning Gives Us Predictions

Three are flagged as fraud  
and the three labels are identical

|                                     |                                     |                                          |                                        |
|-------------------------------------|-------------------------------------|------------------------------------------|----------------------------------------|
| Café Central<br>€4.20<br>legitimate | Supermarket<br>€63.10<br>legitimate | Electronics Online<br>€1,299.00<br>fraud | Petrol Station<br>€78.40<br>legitimate |
| Gift Cards ×12<br>€600.00<br>fraud  | Bookshop<br>€24.90<br>legitimate    | Hotel Booking<br>€410.00<br>fraud        | Pharmacy<br>€12.75<br>legitimate       |

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Would you block all three cards?

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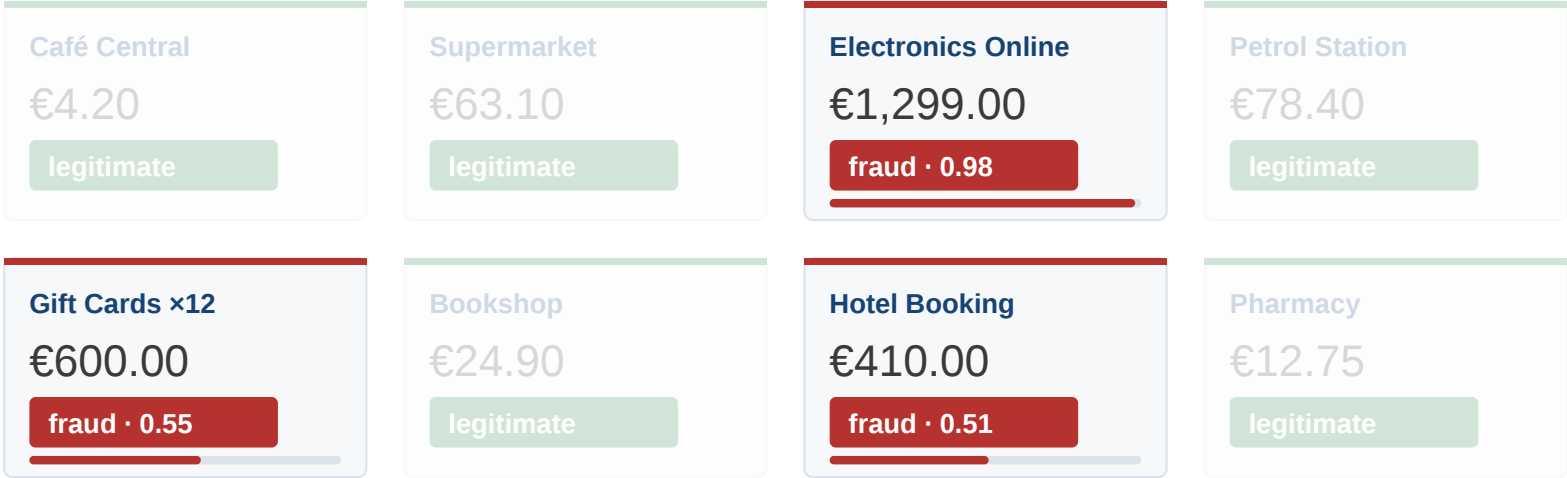
Would you block all three cards?

What would you need to know to treat them differently?

# The Same Batch, With a Number Attached

Now every flagged case carries a number

0.98, 0.55, 0.51 — three very different cases



What can you do now that you could not do a minute ago?



# What a Number Lets Us Do

| The number lets us      | What that means for this batch                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| Compare and rank        | separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud" |
| Weigh consequences      | set the threshold from the cost of a missed fraud against the cost of an annoyed customer                         |
| Defer deliberately      | send 0.55 and 0.51 to a human ask whether more data would move them                                               |
| Combine and communicate | merge evidence from several signals report <i>how sure</i> we are, not only what we think                         |

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**We need to model uncertainty — and we need to be able to reason with it.**

# From Prediction to Managed Uncertainty



And once we want to manage it — **where does the uncertainty actually come from?**

Probabilistic machine learning is not a different way of producing predictions. It is a **language for reasoning about predictions that are not certain.**

# Mini-Exercise

A loan model gives three applicants a **default probability** of **0.05**, **0.35** and **0.45**.

## COST OF ACCEPTING A DEFaulTER

**€8,000**

the loan is not repaid

## COST OF REJECTING A GOOD APPLICANT

**€500**

the margin we would have earned

1. Which applicants does the rule "reject if  $p > 0.5$ " reject?
2. **Weigh each option by how likely its bad outcome is.** What is the expected cost of *accepting* each applicant, and of *rejecting* each?
3. At which probability do the two expected costs balance?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

Weigh each outcome by its probability:  $\mathbb{E}[\text{accept}] = p \cdot 8,000$  against  $\mathbb{E}[\text{reject}] = (1 - p) \cdot 500$ .

|   | $p$  | Accept | Reject | Cheaper by | Decision                  |
|---|------|--------|--------|------------|---------------------------|
| A | 0.05 | €400   | €475   | 1.2×       | accept — <i>only just</i> |
| B | 0.35 | €2,800 | €325   | 8.6×       | reject, clearly           |
| C | 0.45 | €3,600 | €275   | 13×        | reject, overwhelmingly    |

**Break-even:**  $p \cdot 8,000 = (1 - p) \cdot 500 \Rightarrow p^* = \frac{500}{8,500} \approx 0.059$



# What the Threshold Tells Us

## THE INTUITIVE RULE

**reject if  $p > 0.5$**

rejects **nobody** here — every applicant is below it, including one whose default risk is nine times the bank's break-even

## THE RULE THE COSTS IMPLY

**reject if  $p > 0.059$**

because a default costs **16×** what a lost customer does

The threshold comes from the **costs**, not from the probability. 0.5 is only the right cut-off when both mistakes are equally expensive — which they almost never are.

- **B and C are not close calls:** rejecting is 8.6× and 13× cheaper.
- **A is the only borderline case** — €400 against €475, and  $p = 0.05$  sits just under  $p^* = 0.059$ . This is where a second signal is worth buying.

# Takeaway

A prediction without a measure of its uncertainty cannot be compared, weighed against consequences, or handed to a human at the right moment. **Modelling uncertainty and reasoning with it** is what turns a prediction into a decision.

# Where Uncertainty Comes From

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**Guiding Question:** The model says 0.55. Is that because the world is noisy, or because the model has not learned enough — and would more data help?

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

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**Or has the model simply not seen enough cases like this one?**

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**Is the transaction itself ambiguous** — would a perfect model, with all the data in the world, still be unsure?

**Or has the model simply not seen enough cases like this one?**

**Would collecting more data help** — and could *any* amount of data remove the doubt entirely?

# Two Examples

## EXAMPLE A · AN AMBIGUOUS OBSERVATION

Two patients with **identical** records — same age, same symptoms, same test results. One recovers, one does not.

✗ Would more patients like these help?  
Never.

## EXAMPLE B · AN UNFAMILIAR CASE

A house-price model trained on flats of 40–120 m<sup>2</sup> is asked about a **600 m<sup>2</sup> villa**. It has no examples anywhere near it.

✓ Would ten villas in the training data help?  
Completely.

Which uncertainty can be reduced, and which is part of the problem itself?

# The Distinction — Then the Names

|                              | Where it comes from                                | More data? | Our example                            |
|------------------------------|----------------------------------------------------|------------|----------------------------------------|
| Uncertainty in the problem   | the outcome genuinely varies given what we observe | ✗ never    | identical patients, different outcomes |
| Uncertainty in our knowledge | we have not seen enough to pin the model down      | ✓ yes      | the 600 m <sup>2</sup> villa           |

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## Aleatoric

*from alea, dice* — noise in the world

## Epistemic

*from episteme, knowledge* — ignorance in the model

# Why the Distinction Matters

| If the uncertainty is...               | The right response                                                   | The expensive mistake                        |
|----------------------------------------|----------------------------------------------------------------------|----------------------------------------------|
| <b>in our knowledge</b><br>(epistemic) | collect data where the model is thin defer to a human who knows more | accepting it as inherent and never improving |
| <b>in the problem</b><br>(aleatoric)   | model the noise set a policy for the ambiguous cases                 | buying more data that cannot help            |

A model that can tell the two apart can report that it is **out of its depth**, not merely unsure.

A probabilistic model can represent **both** kinds of uncertainty — and can often tell you which one it is facing. That is why the distinction is worth a name.

# Takeaway

Uncertainty has two sources: **noise in the world** and **gaps in our knowledge**. Only the second shrinks with data, and a decision — collect more, defer, act — depends on knowing which one you face.

# Beyond True or False

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**Guiding Question:** A proof either holds or it does not. A learned prediction is never a proof — so what kind of conclusion is it, and how strong is it?

# Three Modes of Reasoning

|                                       | From              | To                                    | What it guarantees                       |
|---------------------------------------|-------------------|---------------------------------------|------------------------------------------|
| <b>Deduction</b> <i>(logic)</i>       | premises          | what <b>must</b> follow               | certain — but adds no new content        |
| <b>Induction</b> <i>(statistics)</i>  | observed patterns | a <b>regularity</b> , a hypothesis    | nothing — content-adding, not guaranteed |
| <b>Abduction</b> <i>(probability)</i> | observations      | the <b>best available explanation</b> | nothing — only comparable against rivals |

Machine learning lives in the bottom two rows.



# The Scientific Reasoning Cycle

The three modes are not alternatives — they take turns.

**Example: gravity.**

## Deduction

From the theory: all objects of equal mass fall at the same rate everywhere on earth.

## Induction

Design an experiment, measure fall times at several locations, compute means and standard deviations.

## Abduction

Times agree at one location but vary between locations.  
**Best explanation:** distance from earth's mass centre.

The cycle turns: the abduced explanation becomes the next theory to deduce from. A data-science project runs the same loop, usually several times.

# Mini-Exercise

A dashboard shows that customers who use the mobile app churn **less** than those who do not.

- 1 What would a purely **inductive** reading conclude?
- 2 Give two different **abductive** explanations.
- 3 What **deduction** would let you distinguish them?

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## INDUCTION

"App use reduces churn" — a regularity read straight off the data.

## ABDUCTION

The app causes loyalty — *or* loyal customers install the app. Same correlation, opposite causal direction.

## DEDUCTION

If the app causes loyalty, randomly prompting a subgroup to install it should lower their churn — an experiment deduced from the hypothesis.

# Beyond True or False

|                        | Classical reasoning       | Probabilistic reasoning           |
|------------------------|---------------------------|-----------------------------------|
| <b>Conclusions</b>     | follow, or do not         | are supported to <i>degrees</i>   |
| <b>Evidence</b>        | premises, taken as given  | may be partial or noisy           |
| <b>Certainty</b>       | the conclusion is certain | the conclusion may stay uncertain |
| <b>New information</b> | can only add premises     | can <i>change</i> what we believe |

Classical logic asks *whether* a conclusion follows. Probabilistic reasoning also asks **how strongly the evidence supports each possible conclusion**.

# What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

- 1 **Represent** — say how much we believe each possibility
- 2 **Combine** — put several uncertain pieces of information together
- 3 **Update** — change our beliefs when new evidence arrives
- 4 **Decide** — act on beliefs that are not certainties

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**Probability provides that language.** The formal material that follows exists because each of these four operations turns out to have exactly one consistent set of rules.

# Takeaway

Machine-learning predictions are inferences from evidence, not derivations from premises.  
**Probability is the calculus of such inferences** — which is why the next lecture is about it, and why it will feel like an answer rather than a prerequisite.

# What Probabilistic ML Enables

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**Guiding Question:** Which questions about a machine-learning system will you be able to answer after this course that you cannot answer today?

# After This Course, You Will Be Able to Ask — and Answer

## ABOUT A PREDICTION

- ? How certain is it, and should I trust it?
- ? When should the model refuse to decide?
- ? What alternatives are plausible?

## ABOUT THE DATA

- ? When would more data help?
- ? When is uncertainty unavoidable?
- ? What does this data actually justify?

## ABOUT LEARNING

- ? How should belief change when evidence arrives?
- ? How do I combine prior knowledge with observations?
- ? How do I compare competing explanations?

## ABOUT THE METHODS

- ? Why do the common methods work, and under which assumptions?
- ? When should I expect them to fail?
- ? What is my model assuming without telling me?

# What Changes Compared to Traditional Machine Learning

## TRADITIONAL ML ASKS

What is the prediction?

## PROBABILISTIC ML ALSO ASKS

- ? **How certain** is it?
- ? **What alternatives** are plausible?
- ? **Why** are we uncertain?
- ? **How does new evidence** change what we believe?
- ? **What decision** follows from this?
- ? **What can and cannot** be inferred from this information?

# Takeaway

Probabilistic machine learning does not replace prediction. It **surrounds every prediction with the questions a decision needs answered** — and gives one language in which all of them can be asked.

# How Machines Learn

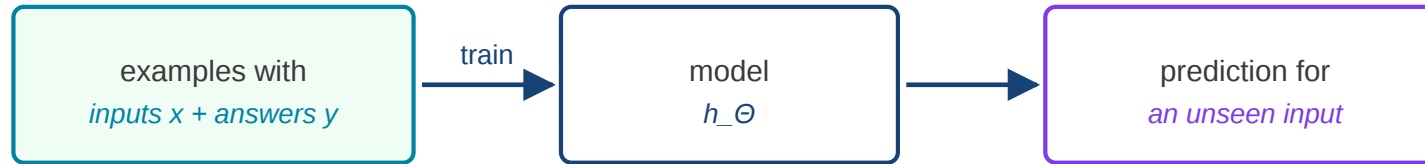
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**Guiding Question:** Nobody writes the rule. So what exactly is a machine given, what does it build, and where does uncertainty get in?

# Learning From Answers

A radiologist has labelled ten thousand scans. A bank has ten years of loans and knows which ones defaulted. In both cases **every example carries its answer**.

Supervised: every training example carries the answer



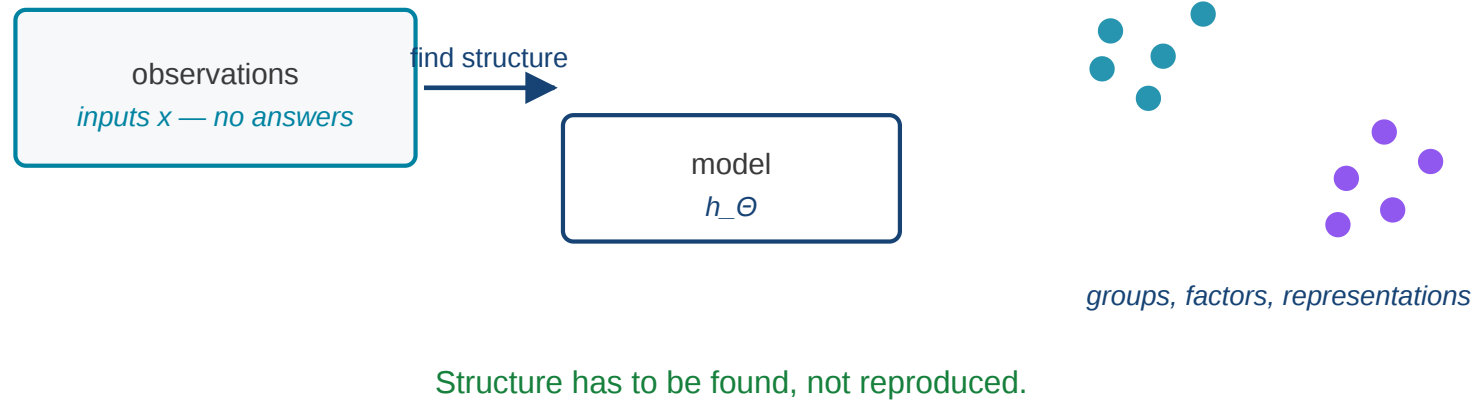
The learner is told the correct output for each case it sees.

What information does the learner receive during training?

# Learning Without Answers

Now the same bank wants to know **what kinds of customer it has**. Nobody has ever written down the list — the categories are what we are looking for.

Unsupervised: no answers — only the observations themselves

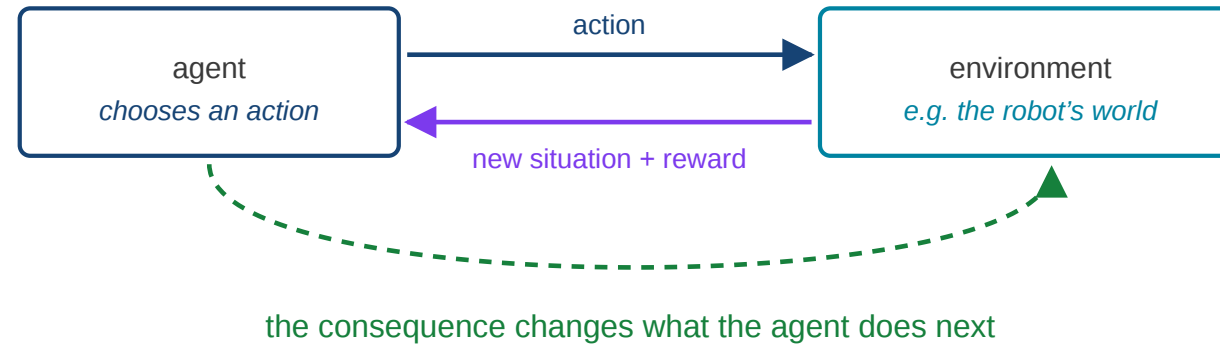


What can we learn when nobody tells us the correct answer?

# Learning by Acting

A robot has to cross a room. Nobody can label the correct motor command for every situation it might find itself in — and it only discovers whether a move was good **after making it**.

Reinforcement: no answer is given — only the consequence of acting



The learner must act to find out whether the action was good.

How can a learner improve when it must act in order to find out whether the action was any good?

# Same Goal, Different Information

Same goal, different information

|               | WHAT IT IS GIVEN         | WHAT FEEDBACK IT GETS  |
|---------------|--------------------------|------------------------|
| Supervised    | inputs + answers         | the answer, every time |
| Unsupervised  | inputs only              | nothing                |
| Reinforcement | the situation it acts in | a reward, after acting |

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The paradigms differ in **what the learner is given** and **what feedback it gets** — not in what we ultimately want from it.

# What Is Common to All Three?

## Nobody wrote the rule

The mapping from situation to answer is **too complex or too implicit** to specify by hand.

## Examples exist

There is data — or experience — in which the desired behaviour is demonstrated, even if it was never described.

So what replaces the rule that nobody wrote?

# Two Ways to Get Behaviour

Both are judged on situations neither of them has seen

## TRADITIONAL PROGRAMMING



*the rules are written down explicitly*

## MACHINE LEARNING



*the rules are inferred from what the environment produced*

# What Actually Changed

## TRADITIONAL PROGRAMMING

The developer's **knowledge becomes rules**. Deduction: rules plus input give the output.

*The assumptions are visible — they are in the code.*

## MACHINE LEARNING

The environment's **data becomes a model**. Induction: examples plus a learning procedure give a rule.

*The assumptions are still there — but now they are in the data, the model family and the objective.*

The rules do not disappear when nobody writes them. They become **harder to inspect** — which is why the rest of this lecture is about making every part of that pipeline explicit.

# What Would It Mean for a Machine to Learn?

Before any definition — three questions you would have to answer about *any* system before you could say it had learned anything:

- ? What is it supposed to get better at?
- ? What does it get to look at while getting better?
- ? How would we know whether it improved?

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## **Task $T$**

what the system should learn  
to do

## **Experience $E$**

what it learns from

## **Performance $P$**

how we decide it improved

# Chess



A program that plays, loses, and plays again — millions of times against itself.

- **Task** — play chess well
- **Experience** — games played against itself, and positions from expert play
- **Performance** — fraction of games won; rating against other engines

# A Large Language Model



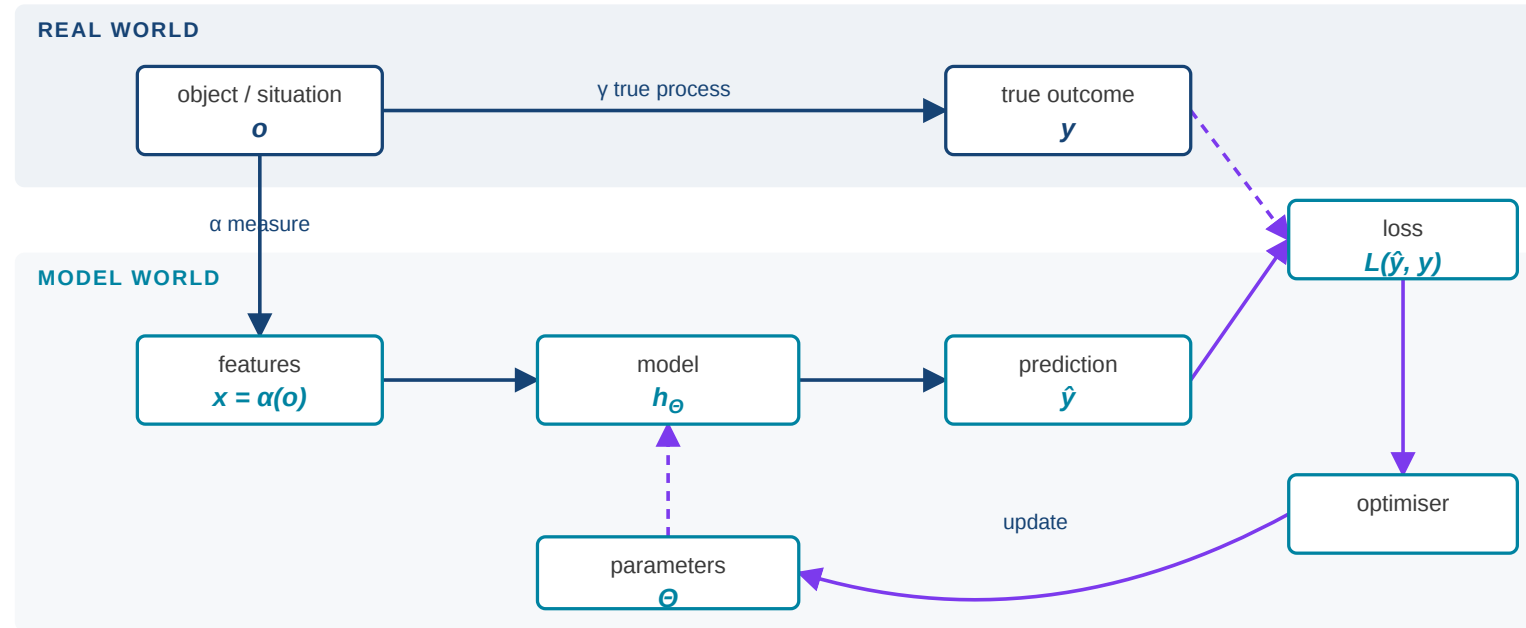
A model shown enormous quantities of text, asked only to continue it.

- **Task** — predict the next word
- **Experience** — a very large corpus of text, then human feedback
- **Performance** — likelihood of held-out text; later, human judgement

Two systems with nothing in common — a board game and a text corpus — fit the **same three-part description**. That is what makes it a definition rather than a description of one program.

# What Would We Ideally Like to Know?

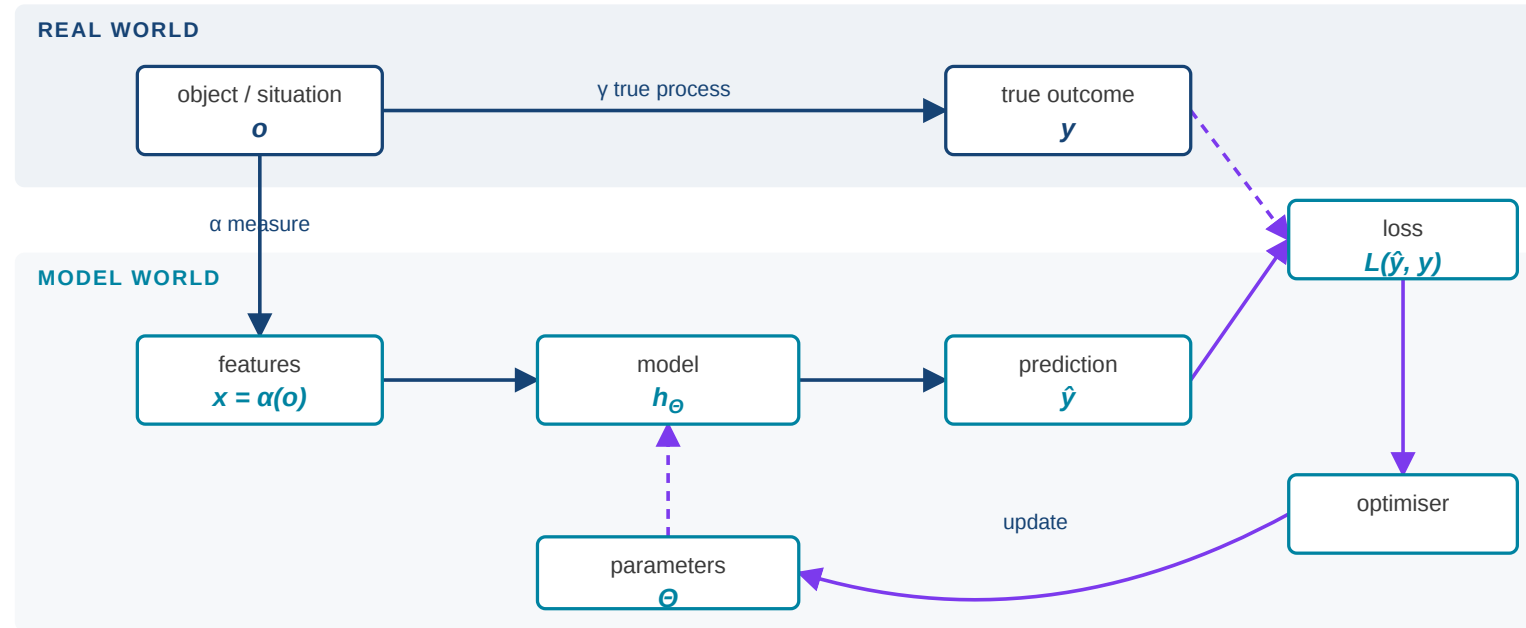
The optimiser changes the parameters — and the training loop closes



There is some real process that turns a situation into an outcome. Why can we not simply write it down?

# From the Real World to a Model of It

The optimiser changes the parameters — and the training loop closes



# Two Worlds, One Correspondence

$$\gamma(o); \rightsquigarrow; h_{\Theta}(\alpha(o)); \equiv; h_{\Theta}(x)$$

$\gamma$

the **real process** — unknown,  
and not ours to write down

$\alpha$

the **representation** — what  
we chose to measure

$h_{\Theta}$

the **model** — our  
approximation, with  
parameters  $\Theta$

We never learn the world. We build a **model world** and approximate the process inside it.

# Closing the Loop

The optimiser changes the parameters — and the training loop closes



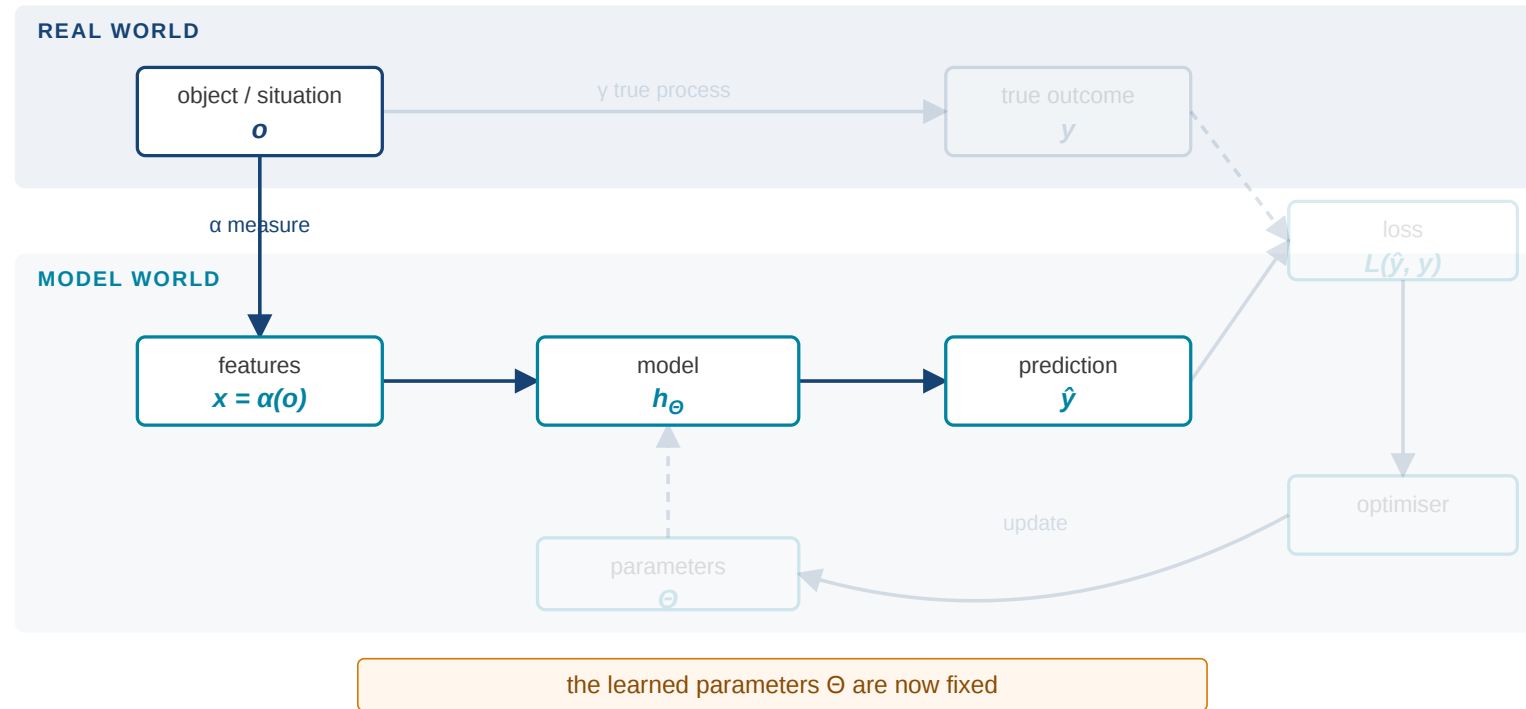
# The Four Choices



The optimiser has settings of its own that are not learned from the data — **hyperparameters**. Choosing them is part of the outer process, not of the loop.

# Training Is Not Using

Inference: a new, unseen situation follows one path — the training machinery is gone



Which of these boxes still exist when the system is actually deployed?

# The Loop Around the Loop

Data, model, loss and optimiser describe **what runs**. They say nothing about how anybody decided on them.

## INNER LOOP · THE LEARNING SYSTEM

data → model → prediction → loss →  
optimiser → updated parameters

*runs thousands of times per minute,  
automatically*

## OUTER LOOP · THE DEVELOPMENT PROCESS

collect → represent → split → train →  
validate → diagnose → revise → evaluate  
→ deploy → monitor

*runs a handful of times per project, and  
every step is a human decision*

# What the Outer Loop Decides

## BEFORE TRAINING

- what to collect, and from where
- how to clean and represent it
- how to split train / validation / test

## WHILE DEVELOPING

- which model family, which hyperparameters
- which errors to look at, and why they happen
- what to change next

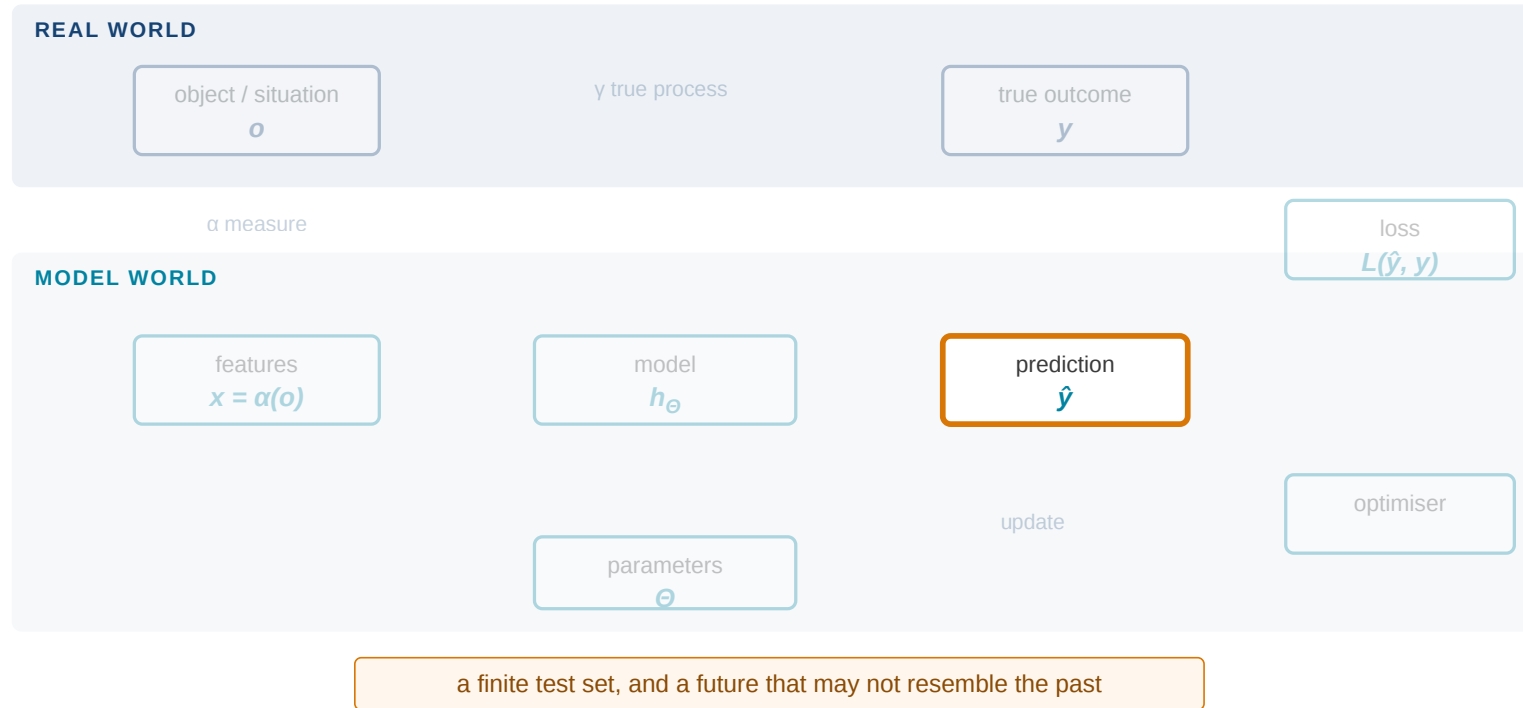
## AFTER TRAINING

- is it good enough, and by whose measure
- does it hold up on data it has never seen
- is it still holding up next month

Machine learning is **experimental**: you make choices, then test whether they worked. Measured performance is *evidence* about future performance — never a guarantee of it.

# The Same System, Read for Uncertainty

In the evaluation



# Two Distinctions Worth Keeping Straight

## UNCERTAINTY *IN* THE LOSS

The quantities entering the loss are uncertain:  $y$  is noisy, the sample is finite. The loss itself is an ordinary function.

## UNCERTAINTY *ABOUT* THE LOSS

Whether this objective represents the task at all. That is a modelling judgement, not a random variable.

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Uncertainty is not a property of the final prediction. It **enters throughout the system** — and probabilistic machine learning is the language for representing and reasoning about it at every one of these places.

# Designing the Learning Problem

Before any model is chosen, four things have to be decided — and each of them is a claim about the world.

## THE TASK

- ? What should the system predict or decide?
- ? What is one unit of prediction?
- ? What counts as success?

## THE DATA

- ? Which observations represent the problem?
- ? How were the examples sampled?
- ? Will future cases look like these?

## THE REPRESENTATION A

## THE TARGET



? What are the inputs, concretely?

? Which features are available — and which are merely convenient?

? What is lost in the measurement?

? What is  $\gamma$ , in words a domain expert would accept?

? Is the label we have the thing we care about, or a proxy for it?

# Designing the System

## MODEL AND OBJECTIVE

- ? What does the model family assume, and how flexible is it?
- ? Does minimising this loss correspond to what we actually want?

## LEARNING

- ? Which procedure searches the parameters?
- ? How sensitive is the result to initialisation and randomness?

# Designing the System — Evaluation

## EVALUATION AND GENERALISATION

- ? Which metric matters, and which data is held out?
- ? How uncertain is the estimate of performance?
- ? What happens when the environment changes?

## UNCERTAINTY AND USE

- ? Which uncertainty must be represented explicitly?
- ? Is the prediction informing a decision — and what do the errors cost?

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## UNCERTAINTY AND USE

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These are the questions the rest of the course answers. Each lecture takes one or two of them and makes them precise.

# The Road Ahead

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**Guiding Question:** Now that the motivation is in place — which lecture answers which of your questions?

# The Road Ahead

Each lecture answers one of the questions you have just met.

**02**

**Representing uncertainty** *How do we describe uncertainty mathematically?*

Probability, distributions, and how to summarise data.

**03**

**Learning from evidence** *Why do the classical methods work — and what do they assume?*

Linear and logistic regression, re-read as probabilistic models.

**04**

**Combining information** *How do uncertain quantities relate — and when can we treat them separately?*

Naive Bayes, independence, and the graphical view.

**05**

**Structured models** *How do we write down a large, structured set of beliefs?*

Bayesian networks, fitting them, and drawing them.

**06**

**Judging models** *What is the quantity we actually want to minimise — and what did we get instead?*

Evaluation and empirical risk minimisation.

**07**

**Deciding and refining** *How do we keep uncertainty all the way to the decision?*

Mixtures, Bayesian regression, and what comes after.

# The Programme, in Three Moves

Every method in this course is the same three moves, in the language you are about to learn:

## Model the joint

Write down what the world would have to be like to produce what we observed — the data *and* the unknowns together.

## Condition on the data

Inference is conditioning: fix what was observed, and let the rules of probability say what follows for the rest.

## Read off answers

Predictions, uncertainties and decisions are all quantities computed from that one conditioned model.

Every one of these words — joint, condition, inference — gets a precise meaning in Lecture 02. They appear here only so that you recognise them when they return.